From *Habitus* to *Systema*:

Keckermann and the Pedagogical Origins of Cognitive Systematicity

Abstract

This article argues that the application of *systema* to an organized body of knowledge emerged around 1600 not first as a metaphysical or epistemological ideal, but as a pedagogical solution to problems generated by confessional fracture and educational scarcity in Reformed Protestant Europe. Focusing on Ramism, Bartholomäus Keckermann, and Johann Heinrich Alsted, it shows how the new system externalized cognitive order, transferring work once performed by cultivated *habitus* and institutional authority into visible, reproducible structures of exposition. The early modern "system" thus began life as a teaching machine, and only later became the model for more ambitious claims about rationality, certainty, and impersonal authority.

> "Philosophy, as it exists in the mind, is a *habitus*; as it exists outside the mind, it is a *systema*."
>
> – Johann Heinrich Alsted

The word *system* is so thoroughly assimilated into the vocabulary of modern thought that it can be difficult to hear it as a word with an origin: a term that was, at a specific historical moment, seized upon by identifiable thinkers grappling with identifiable pressures. Yet the application of *systema* to an organized body of knowledge is a distinctly early modern invention. Why did European thinkers come to demand that knowledge form an articulated system rather than a loose aggregate of truths? This article provides one answer. It traces the concept of a cognitive *systema* to the late-sixteenth-century Reformed Protestant universities of northern Europe, and it argues that what we find there is something the later philosophical tradition has largely obscured: the system was born as a practical technology before it became a philosophical ideal. It first arose as a way of making doctrine and learning surveyable, reproducible, and transmissible under conditions in which neither personal apprenticeship nor inherited institutional authority could be relied upon. In this first historical setting, the primary rationale for systematization was pedagogical. Yet in answering that pedagogical need, the *systema* also initiated a broader shift whose consequences reached far beyond the classroom: more and more of the work once performed by formed persons and authoritative institutions was transferred into articulated structures. Recovering the circumstances of this invention is therefore the first step toward understanding both the later prestige of the systematicity ideal and the peculiar hazards that would accompany it.

In Greek antiquity, *systēma* (from *syn-histēmi*, “to make to stand together”) designated a composite whole of a quite general kind: a flock of animals, a military formation, a composite political unit, a musical scale. The *systēma teleion* of Aristoxenus was the complete tonal system of Greek music theory; the *systema mundi* of the Stoic cosmologists was the rationally ordered cosmos itself, heaven and earth and everything contained within them, held together by divine *logos* (Rescher 2006, 90). The term continued in use throughout classical antiquity in this broad ordinary sense: something joined together, a connected or composite whole.

The Stoic usage deserves particular attention, because it harboured an ambiguity that would prove significant. The Stoics conceived of the cosmos as a providentially ordered totality governed by rational principles, in which physics, logic, and ethics formed an organic unity, comparable to a living being whose bones (logic), flesh (ethics), and soul (physics) supported one another. This was the idea of a *rational* whole whose parts were held together by intelligible connections. And yet *systēma* remained, for the Stoics, tied to ontological rather than epistemological systematicity: it described the organized totality of the world – how *things* stand together, rather than how *beliefs* about things do. The semantic leap from *systema mundi* to *systema doctrinae*, from the rational order of the cosmos to the rational order of a textbook, would have to wait more than a millennium.

This is not to say that the Greeks lacked practices of systematic deductive organization. Euclid’s *Elements*, composed around 300 BCE, provided the most enduring paradigm of knowledge organized into axioms, definitions, and rigorous derivations, and it would later become the inspiration for philosophers who sought to cast all of knowledge in a deductive mould. Yet the Greeks did not describe Euclid’s achievement as a *systēma*. The term belonged to cosmology and music theory; the deductive ordering of geometry was a separate intellectual achievement, one whose enormous potential for the organization of knowledge would be recognized only when early modern thinkers retrospectively adopted it as the model of what a system should look like. The adoption of the deductive model is a separate story from the one told here. The present article tells a different story: the rise of an organized, externally displayable architecture of learning.

When the semantic leap from *systema mundi* to *systema doctrinae* finally occurred, it occurred in a setting that differed in many respects quite profoundly from the world of Euclid. Prior to the late 1590s, organized presentations of theological or philosophical learning employed terms such as *summa*, *loci communes*, *corpus doctrinae*, *compendium*, and *institutio*, but never *systema* (Ritschl 1906, 16, 25–30). By 1906, however, Ritschl counts at least 90 books with "system" in their title – 3.4 books per year on average (1906, 26).

The term's migration from physical composite to principled cognitive architecture was accomplished, almost single-handedly, by a Calvinist pedagogue working in a Baltic port city. His name was Bartholomäus Keckermann, and his *Systema logicae* of 1600 launched a transformation in how Europeans conceived of knowledge. To understand why Keckermann was driven to systematize knowledge in this way, and why he reached back to the Stoic term for an ordered cosmos to describe an organized body of doctrine, we must first understand what this new ideal of cognitive systematicity replaced.

1. The World Before the Systematization of Knowledge

For roughly five centuries, from the high Middle Ages through the early Reformation, European intellectual life aspired to a form of cognitive organization that was highly sophisticated, yet profoundly different from the striving for systematicity. Scholastic knowledge fundamentally rested on two foundations. The first was ontological: the "great chain of being," the hierarchical ordering of all creation from God through the angels and human beings down to brute matter, in which every entity occupied a determinate place and every discipline studied some portion of a single divine arrangement (Lovejoy 1936, 59–66). The second was institutional: the authority of the Catholic Church, whose Magisterium served as the ultimate arbiter of doctrinal coherence. Where apparent conflicts arose between Scripture, the Church Fathers, and Aristotle, the Church possessed the standing to settle the dispute. Cognitive order, on this dispensation, was sustained by ontological hierarchy and institutional authority, and it did not require articulated structure to do the work of holding things together.

The organizing ideal of knowledge was dialectical disputation. Knowledge, in this context, advanced through the disciplined confrontation of competing authorities. As Walter Ong has emphasized, intellectual activity in the ancient and medieval world was conceived primarily in auditory and conversational terms: the Socratic method essentially consisted of oral dialogue; Aristotle's categories, his *predicaments*, were at root things *said of* a subject – they were vocal assertions, or accusations brought against a subject, and his notion of predication was grounded in the act of *saying* (Ong 1956, 224; 1958, 53). One arrived at knowledge by hearing it from a teacher, conversing about it, and being enculturated into a tradition of disputation. The authority of a claim rested in significant part on the authority of the person or tradition from which it issued, and learning was accordingly bound up with the prolonged presence of recognized masters.

This also meant that to learn philosophy was primarily to develop a *habitus*: an acquired intellectual disposition cultivated through years and years of study and dialogue (Freedman 1999, 92–93; Hotson 2007, 45–46). This involved sharpening one's judgment, accumulating experience, developing an acute sensitivity to context, and learning how to weigh competing considerations. On this still thoroughly Aristotelian conception of pedagogy, knowledge was instilled in *someone*: *habitus* was embodied in persons, and it was understood to take a very long time to acquire. A traditional philosophical education demanded twelve to sixteen years of study before mastery was attained. One did not join the ranks of the *sapientes* overnight.

The paradigmatic expression of this organization of knowledge through dialectical disputation was the *Summa* – above all, Thomas Aquinas's *Summa Theologiae* (1265–1274), that monument of Scholastic method which was also the standard textbook of advanced theological education. The *Summa* was structured in terms of dialectical disputation: a topic was broached as a *quaestio* (question), opposing arguments were marshalled under *videtur quod* ("it seems that"), and an authoritative counter-statement was provided in the *sed contra* ("but against this"). The author's own resolution was then argued in the *respondeo* ("I answer that"). Crucially, however, it was followed by the *ad argumenta* ("replies to the objections"), which accomplish something highly characteristic of this way of organizing knowledge: as Anthony Kenny observes, they "go a long way to accommodate the opposite view" (Kenny 1993, 22). The very act of forcing oneself to write out the objections in their strongest form and do them justice in one's replies forces one to elevate or expand one's view sufficiently to accommodate both sides of the issue.

This disputational organization allowed Aquinas to preserve the tension between two sides of an argument. Where the later demand for perfect consistency and coherence would encourage one to flatten paradoxes and deny the existence of irreducible conflict, the *Summa* thrived on accommodating both sides of a question and exploiting the productive tensions between them. The positions at which Aquinas arrives are judgments refined through confrontation with competing considerations. The authority of the conclusion rests in part on the visible process by which it was reached; readers understand the position because they have seen it tested and were brought to appreciate the force of countervailing considerations.

The *quaestio* format of the *Summa* thus tied knowledge to spoken disputation, to the give-and-take of voices in a room, while the notion of *habitus* tied it to a particular person having undergone a long formation. Knowledge, on this model, was embedded in discourse and persons, in a world still fundamentally shaped by the primacy of the spoken word. It could certainly be ordered, often with extraordinary rigour; but it was not yet optimized for rapid surveyability, for uniform replication across many learners, or for relative independence from the master who transmitted it.

This disputational conception was supported by the humanist pedagogy of *copia* (abundance), articulated by Erasmus in his *De copia verborum ac rerum* (1512) and inspired by Cicero and Boethius, whereby one should perfect one's rhetoric by building up a storehouse of material in commonplace books. The aim was to help the speaker adopt a wide variety of viewpoints and generate a great variety of arguments, examples, and formulations. Commonplace books were so ubiquitous that, as Ann Blair has shown, they constituted the default method of learned reading and writing across Europe for centuries (Blair 2010, 62–83). Because commonplace books organized material by topic rather than by principle, however, they facilitated retrieval and recombination rather than principled derivation. They consisted of loose collections of excerpts and observations organized under topical headings, but otherwise remained radically unsystematic.

Yet it was out of these "commonplaces" that a significant intermediate step on the path to *systema* emerged: the *loci communes* of early Protestant theology reconceived the collection of commonplaces as a method for mapping out the structure of reality itself. Topics such as cause, effect, genus, species, similarity, and difference were treated as classes of things that map the real relationships among entities, providing a topical map through which any subject matter could be navigated (Burton 2024, 27–29). Melanchthon, in his *Loci Communes* of 1521, presented theology as a series of topics arranged to mirror the intrinsic structure of Scripture. The *loci communes* thus occupied a position midway between the disputational depth of the *Summa* and the architectural rigour of the later *systema*: it organized knowledge topically rather than deductively, but it aspired to track the objective order of its subject matter instead of merely storing material for retrieval. Calvin's *Institutes*, as Richard Muller has shown, evolved from a catechism into a pioneering system of Reformed theology under the influence of precisely this topical-methodological tradition, in which the right arrangement of theological *loci* was understood to reflect "the right order of teaching" (Muller 2003, I:177–180). It was this tradition that Keckermann inherited and transformed.

This pre-systematic order thus had its own forms of cohesion, indeed its own depth. Bernard Williams's observation about the "intuitive condition", in which convictions and commitments form a dynamic and dialectical fabric rather than a unified but static structure, captures something of what this older intellectual world was like. The *quaestio* format exemplified this condition at the level of method: it rendered disagreement productive, holding competing authorities in tension rather than dissolving that tension through derivation from first principles. The humanist tradition of *copia* cultivated plurality, enriching the speaker's and reader's resources for approaching a question from multiple angles. And the conception of knowledge as *habitus* prized precisely those qualities that later system-builders would work to make less indispensable: personal judgment, accumulated experience, sensitivity to context, and the capacity to weigh competing considerations without reducing them to a common currency.

Even the topical tradition, which came closer to the later *systema* in aspiring to track an objective order, remained committed to a form of intellectual navigation that depended on the reader's own capacity to identify the right topics and trace the right connections. It provided a map, but one that still required a skilled traveller. A Keckermann-style *systema* covering the same material as a *Summa* would state the doctrine, provide the definition, and move on. The dialectical depth – the sense that the position has been tested against alternatives and shaped through engagement with them – would be attenuated. What would eventually be gained by the transition to *systema* was real, as the rest of this article will show: speed, surveyability, reproducibility, and a form of cognitive order less dependent on the qualities of particular teachers. But something was also relinquished.

The question, then, is why anyone found the proposal to replace this order with a *systema* compelling. The answer – like so many things in the early modern period – begins with the Reformation.

2. The Loss of Institutional Authority and Educational Infrastructure

The Protestant Reformation shattered the two foundations on which Scholastic cognitive order had rested. It disrupted the Church's monopoly on doctrinal authority and the institutional network through which that authority had been reproduced in learned persons. What emerged in its place was a new problem of cognitive organization. If a body of doctrine could no longer rely, in the old way, on the authority of a universally recognized Church, and if the educational institutions that had formed its guardians were themselves weakened, dispersed, or abolished, then order had to be secured differently. Consequently, as Richard Muller has documented, the Reformers "were forced by polemical and practical necessity to develop theological systems and schools designed for creating and maintaining the Protestant movement" (Muller 2003, I:76).

The sixteenth century thus generated two pressures that were analytically distinct, though historically intertwined. The first was a pressure toward replacing inherited authority with articulated doctrinal order. The second was a pressure toward making that order transmissible with new speed, uniformity, and economy.

The first pressure arose from confessional fracture itself. Before the Reformation, the Church could serve, at least in principle, as the final arbiter of doctrinal disagreement. Its authority did not eliminate controversy, but it provided a standing institution within which controversy could be contained. Protestantism, by contrast, weakened precisely that kind of external guarantor. Once multiple confessions claimed access to scriptural truth without acknowledging a single magisterial tribunal, the burden of authority shifted. Doctrine could no longer appear merely as what an institution authoritatively maintained; it increasingly had to present itself as an ordered whole whose internal articulation made its rightness visible, or at least its intelligibility defensible. The thought that order might reside in structure rather than solely in office or inheritance received, in this way, a powerful historical stimulus. The demand for internal structural criteria was simultaneously a demand for a new locus of intellectual authority, one vested in the architecture of a system rather than in the office of an arbiter.

The second pressure was pedagogical. The Reformation dismantled and reconfigured much of the educational apparatus that had sustained the old order. Monasteries, cathedral schools, and established teaching networks were dissolved or transformed across much of Protestant Europe.

If the new confessions were to survive, they had to train pastors, schoolmasters, and administrators rapidly and at scale. They needed ways of compressing doctrine into forms that could be taught efficiently to relative novices, disseminated across dispersed institutions, and reproduced with a reliability less dependent on the exceptional gifts of individual teachers. In this setting, the attractions of articulated, surveyable, portable forms of knowledge became evident. Protestant universities and gymnasia, many of them newly founded, faced the task of transmitting complex bodies of knowledge efficiently and reliably, to students who could no longer rely on the traditional Catholic infrastructure of training. This pedagogical emergency demanded forms of knowledge that could operate with less dependence on prolonged formation, on the personal authority of the master, and on the institutional settings in which learning had traditionally taken place. It demanded forms in which more of the cognitive work was performed by the arrangement of the material itself.

It is important not to overstate the novelty of either pressure. Medieval theology had long valued order, and every educational regime has practical constraints. But the conjunction of confessional fragmentation, institutional disruption, and print-based dissemination altered the balance of virtues that a body of doctrine was now expected to possess. Under these conditions, the older ideal of *habitus* – slowly cultivated, dialogical, person-bound – began to appear insufficient on its own. It presupposed stable institutions, extended time, and reliable chains of transmission. Protestants could and did continue to prize learned judgment; but they also needed forms of doctrinal organization that could do more of the work externally. The demand was for bodies of learning that could be seen, taught, traversed, and reproduced with greater economy.

This is the point at which the pedagogical rationale for systematicity first becomes historically explicit. The issue was not yet, at least not primarily, how to derive certainty from indubitable first principles, nor how to ground political legitimacy in publicly shareable norms. The immediate question was how to preserve and transmit a body of teaching under new conditions of scarcity and instability. Systematization first answered a problem of reproduction. It arose as a machine for teaching before it became a machine for proving or governing.

The language of "machine" is not meant to insinuate that sixteenth-century pedagogues imagined themselves building literal cognitive automata. But the cognitive virtues they aspired to were in effect the virtues of a pedagogical machine: characteristics that made knowledge surveyable, reproducible, and relatively less dependent on the tacit mastery of formed individuals. Some of the work previously borne by the cultivated person had to be taken over by the structure itself. Order needed to stand there on the page.

No movement more vividly embodied this transfer from person-bound formation to externalized arrangement than Ramism.

3. Ramism and the Visual Reorganization of Learning

Petrus Ramus (1515–1572), the controversial French humanist logician, became one of the most influential educational reformers of the later sixteenth century. His followers spread his methods across Protestant northern Europe, especially in Reformed contexts eager for pedagogical simplification and curricular reorganization. Ramism is often remembered for its hostility to Aristotelian Scholasticism and its fondness for dichotomous division. But its deeper significance for the history of systematicity lies in the way it reconceived knowledge as something that could be rendered spatially surveyable and procedurally manageable.

Ramus criticized the inherited Scholastic curriculum for its prolixity, subtlety, and dependence on disputational complexity. In its place, he proposed a method that would proceed by successive binary divisions from the more general to the more particular. Knowledge was to be arranged in trees, charts, and ordered schemata that displayed relations of inclusion and subordination at a glance. The aspiration was pedagogical clarity: to present an art or discipline in such a way that its principal divisions could be quickly grasped, memorized, and taught. Ramus's insistence that every discipline could be reduced to clear definitions and systematic dichotomous divisions, displayed in branching tables that made the structure of knowledge spatially visible and immediately graspable, offered Protestant educators precisely the kind of scalable didactic technology their predicament demanded. Where a traditional philosophical education demanded twelve to sixteen years of study before mastery was attained, Ramist textbooks promised competence in months.

The principal pedagogical innovation that promised to fuel these gains in efficiency was the Ramist dichotomy chart or "Ramist tree," which divided a subject into two branches, and those branches into two more. With their rigid, clean, and pedagogical look, such dichotomous diagrams were designed to help students memorize complex subjects by breaking them down into manageable branching pairs. In the sixteenth and seventeenth centuries, these diagrams were a significant departure from established practice, transposing dense text into a spatial layout that made it easier for readers to see the logic and structure of a subject at a single glance. Where Aristotle's dialectic retained its connections to speech and disputation, Ramus severed those connections with what Ong calls "savage rigour": the topics were abstracted from rhetoric, judgment was replaced by spatial "arrangement" (*dispositio*), and the oral, interpersonal dimension of knowledge was displaced by a visual, diagrammatic one (Ong 1958, 280–282, 287). Words, in Ramist textbooks, were deployed in spatial relationships, converting knowledge from something *heard* into something *seen*, arranged on the page as a thing to be surveyed (Ong 1958, 89–91). Learning became less bound up with hearing the personal authority of a teacher and more bound up with inspecting the impersonal authority of a structure.

Here one can already see the formal virtues that would later be associated with cognitive system-building more generally. Ramist method prized surveyability: a discipline should be visible as a whole. It prized decomposability: a subject should be divisible into orderly, hierarchically related parts. It prized rule-guided traversal: one should be able to move through the subject by following explicit divisions rather than relying at every step on tacit judgment or the teacher's improvisational skill. And it prized a kind of portability or generalizability: the same formal scheme could, in principle, be applied to many domains.

This was not yet the mature early modern *systema*. Ramism was often too rigid, too reductive, and too suspicious of the texture of real inquiry to furnish a durable model on its own. But it made visible, perhaps for the first time in early modern pedagogy, the thought that knowledge could be reorganized so as to become externally graspable in a uniform visual architecture. What had previously been embedded in processes of conversation, commentary, and prolonged exercise was now increasingly laid out as a structure to be inspected.

Walter Ong captured something essential about this transformation when he argued that Ramism marked a shift from a more oral, agonistic culture of learning toward a more visual and spatial one (Ong 1958). Ramism, in Ong's analysis, was "a cluster of mental habits ... specializing in certain kinds of concepts, based on simple spatial models, for conceiving of the mental and communicational processes and, by implication, of the extramental world" (Ong 1958, 8). In the older tradition, mastery was displayed in the capacity to navigate arguments in real time, to respond, distinguish, and judge. In the Ramist classroom, increasing importance attached to the capacity of a scheme itself to carry order visibly. The chart became, in a modest but significant sense, a cognitive instrument.

The visual form of the Ramist tree is especially revealing. Rather than letting insight grow out of a dialectical exchange, it prescribes a route through knowledge. The student is not asked primarily to dwell on tensions or cultivate a sensitivity to the force of competing arguments of rival authorities, but to traverse a pre-arranged set of divisions. The scheme therefore itself performs some of the cognitive labour. It guides attention, determines sequence, and renders relations explicit in advance. In that respect, the Ramist diagram may be read as one of the first pedagogical artifacts in which thought begins to take on a distinctly machine-like profile: a free-standing and orderly architecture, instantly surveyable and algorithmically traversable, with less dependence on the dialectical gifts of a master.

The very starkness that made Ramist method attractive to reformers also made it vulnerable to criticism. Where the old disputational method cultivated judgment through engagement with opposition, Ramist method risked replacing the capacity for balanced judgment with a Procrustean tendency to force every issue into simple dichotomies. Critics complained that Ramist dichotomies hollowed out the content of disciplines, purchasing pedagogical convenience at the cost of philosophical substance (Hotson 2007, 131–132). For theologians engaged in sophisticated confessional polemic, this posed a real problem: Ramus's truncated treatment of logic stripped away precisely the metaphysical resources needed to defend theological orthodoxy against attacks (Burton 2024, 225–226). By 1599, Keckermann summed up a broad consensus in blaming Ramus for what he called "one of the calamities of our time": the fact that the fundamental disciplines preparatory to all higher learning were “now seldom seen among students” (Hotson 2007, 144). The students arrived knowing the branching tables, but they struggled to construct a syllogism or detect a fallacy. What was needed was a synthesis: a method that preserved the Ramist virtues of clarity, surveyability, and efficiency while restoring the Aristotelian substance that Ramism had stripped away.

This synthesis was Bartholomäus Keckermann’s achievement. Before turning to him, however, it is worth noting the broader significance of the Ramism that preceded him. In transforming classroom methods, Ramism transformed expectations about what a body of learning ought to be. A discipline was increasingly expected to be displayable in a compact architecture, to have a visible order that could be apprehended in overview, and to be teachable by means of explicitly articulated divisions. This was still a pedagogical expectation before it became an epistemological one. The demand was that doctrine be teachable and navigable. But once that demand had been established, the idea that order should be carried by articulated form had acquired a new plausibility. And with that, a path had opened towards more ambitious uses of structure.

The importance of Ramism lies in what it rendered imaginable. The question it posed to the next generation was whether knowledge could be made at once more intelligible in form and more serious in substance. Keckermann’s *systema* was an answer to that question. It transformed the pedagogical reorganization of doctrine into a more durable architecture of knowledge.

4. Keckermann’s Innovation

Bartholomäus Keckermann (1571–1609) was born in Danzig (now Gdańsk, Poland) and educated in the Reformed intellectual world shaped by Ramism, Aristotelianism, confessional conflict, and pedagogical urgency. He taught first in Heidelberg and then at the Academic Gymnasium in his native city, where he produced, in a brief life, a large number of textbooks spanning logic, ethics, physics, rhetoric, theology, and metaphysics. He produced what has been called a "semi-Ramist Aristotelianism": a synthesis that conceded to the Ramist demand for strict formal organization while insisting on the substantive richness of the Aristotelian tradition (Hotson 2007, 145–146, 149). In 1600, Keckermann published his *Systema logicae*, the first work to apply the term "systema" to a body of learning in the now-familiar sense. He thereby gave a name to a new way of conceiving what an organized discipline should be. And his *Systema logicae* was followed in rapid succession by a *Systema SS. theologiae* (1602), a *Systema disciplinae politicae* (1607), and a posthumous *Systema physicum* (1610); his collected works were published as the *Systema systematum* (1613–14), the "system of systems."[1]

The novelty can be obscured if one interprets the innovation too lexically. Bodies of doctrine had, of course, long been ordered. Medieval theologians and jurists had arranged material with great sophistication; humanists and reformers had experimented with topics, methods, compendia, and institutions. What Keckermann introduced was a more explicit conception of a discipline as a systematically integrated whole whose parts are arranged according to a principled architecture that can be displayed in a form suited to reliable transmission. His *systema* aspired to do more than offer a digest or a didactic depiction. It aimed to display the inner structure of the discipline itself.

[1]Keckermann's student and editor Clemens Timpler (1563–1624) then published his *Metaphysicae systema methodicum* (1604), formulating normative principles for building any system whatsoever and thereby effectively providing a meta-system: a system for building systems (Secundant 2016b, 41–56). In 1610, the Ramist educator Johann Heinrich Alsted (1588–1638) also published his *Systema mnemonicum duplex* at Herborn.

This is why Keckermann occupies such an important place in the genealogy of cognitive systematicity: he turns the pedagogical need for surveyable and reproducible order into a general model of cognition. Under his hand, the educational problem of how to teach a body of doctrine efficiently begins to yield a broader ideal of what a body of doctrine ought to look like.

Keckermann inherited from Ramism the conviction that knowledge had to be rendered methodical, orderly, and accessible to the learner through visible arrangement.[2] Keckermann's own threefold organizational template of *praecognita* (principles of a discipline), *systemata* (their methodical ordering), and *gymnasia* (their practical application) was itself an expansion and modification of Ramus's basic methodical scheme (Hotson 2007, 148–151, 157–165; Burton 2024, 232–233). And when Keckermann's devoted students, working under Johann Heinrich Alsted's editorial direction, undertook the posthumous project of compiling the *Systema systematum*, they found it remarkably easy to resolve all of Keckermann's works into the dichotomous charts characteristic of Ramist presentation. The Ramist skeleton was still there underneath the Aristotelian flesh.

But he also recognized the defects of Ramism when pressed too far. Dichotomous division could impose a schematic neatness at the cost of fidelity to the subject matter, producing structures that might be easy to memorize, yet too thin to sustain serious learning. Keckermann sought to preserve the pedagogical gains of the Ramist reorganization while restoring much of the conceptual density and disciplinary seriousness that Ramism had threatened to evacuate. For instance, Keckermann's engagement with Jacopo Zabarella, the Paduan Aristotelian whose

[2]Burton has shown that Keckermann's entire education marked him as a "Philippo-Ramist": the Danzig Gymnasium where he studied through 1590 was a stronghold of the Melanchthonian-Ramist synthesis, and his subsequent trajectory through the Universities of Wittenberg and Leipzig brought him into contact with Zabarellan analytical method without dislodging his formative Ramist commitments (Burton 2024, 217–219).

methodological writings were reshaping logic across Protestant Germany, equipped Keckermann to go significantly beyond Ramus in his twofold division of method into the *synthetic*, which retained the Ramist order from universals to particulars, and the *analytic*, which proceeded from particulars to universals (Keckermann, *Praecognita logica*, 150–51). This allowed Keckermann to resist the Ramist assumption that there was a single method for all arts and sciences (Burton 2024, 224–226). Keckermann recognized that different disciplines needed to be treated according to their distinctive character. But he preserved the Ramist demand for a perspicuous, methodically organized presentation of each.

Keckermann's choice of title for *Systema logicae* (1600) marks the moment at which the Stoic-cosmological term is deliberately appropriated and extended to the cognitive domain. Just as the Stoic *systema mundi* designated a cosmos held together by rational principles, Keckermann's *systema logicae* designated a body of doctrine held together by logical structure. To call a textbook a *systema* was to make a substantial philosophical claim, namely that the discipline it presented possessed an internal rational order of a particular kind: that its parts were connected by principled relationships, and that the whole could be grasped through methodical exposition rather than through prolonged immersion.

Because of its roots in cosmology, the notion of a system of knowledge evoked the picture of knowledge as a self-contained, organized whole revolving around a centre – its first principles – in a manner analogous to the Copernican spheres revolving around the sun. The very notion that a philosophy could constitute a *system* – something that, in Ong's vivid phrase, "whirled dazzlingly around a center in the mind like the Copernican spheres around the sun" (Ong 1956, 237) – was itself a product of the Copernican revolution.

Before 1600, philosophical positions were differentiated as different *viae* – the *via Thomae*, the *via nominalium* – or as the *opiniones* of different persons. People thought in terms of different persons and their approaches, their "ways," rather than in terms of self-contained intellectual architectures. Thinkers who described their philosophical position as a "system" were claiming for it a kind of structural completeness and internal coherence that the older language of *via* (a "way" or method) or *opinio* (an "opinion" or approach) had never implied. "It was comforting," Ong observes, "to think of oneself, or of one's enemy, as possessing a philosophical 'system'," because it "could cover intellectual situations of which one knew really very little," making the very looseness of the notion of system "one of its greatest recommendations" (Ong 1956, 237).

Keckermann's pedagogical ambition was to give students "an encyclopaedic education within three years": the first year devoted to logic and physics, the second to mathematics and metaphysics, the third to ethics, economics, and politics (Hotson 2007, 154). This was conceivable only if the entire philosophical curriculum could be compressed into the form of systems. It required self-contained and methodically organized presentations in which the surveyable structure took over some of the cognitive work previously performed by apprenticeship and accumulated experience. The systematic textbook became, in effect, a teaching machine: a device for producing competent thinkers by exposing them to the right sequence of definitions, divisions, and derivations, regardless of whether a master was present to guide the process. More of the work was now done by the arrangement of the material: by the definitions that fixed meanings, the divisions that displayed relations, the ordered sequences that guided the reader's progress through the discipline. Less was left to the slow, tacit acquisition of mastery through prolonged apprenticeship. The redistribution of cognitive labour from formed persons to articulated structure is what makes the systematic textbook a machine in the sense that matters here: its virtues are the virtues of a mechanism – reproducibility, inspectability, operator-independence – even as it remains, of course, a thing made of paper and ink rather than of gears and springs.

With Keckermann's original application of the term "systema" to the cognitive domain, the pedagogical rationale for systematization appears in its clearest form. The *systema* responds to the immense pressure on teachers to deliver large bodies of knowledge to expanding student populations in highly compressed timeframes (Hotson 2007, 15–19, 82–87). A body of knowledge must become teachable in a way that is efficient, robust against variation in teachers, and traversable for relative novices without much supervision. Textbooks must, in short, become more like machines for transmitting knowledge.

Keckermann's *Systema logicae* marks the beginning of a shift from *habitus* as embodied knowledge to *systema* as a free-standing cognitive architecture dissociated from individual knowers. Philosophy becomes something that can be written down completely and transmitted through textbooks, without reference to cultivated virtues of the philosopher.

This is made explicit by Keckermann's student and editor, Johann Heinrich Alsted (1588–1638). Philosophy "as viewed in the mind," Alsted wrote, "is a *habitus*, as viewed outside the mind, a *systema*" (Alsted, *Cursus philosophici encyclopaedia*, 1620, I, col. 1).[3] The formulation registers how the older conception of knowledge as a set of personal dispositions acquired through formation and dialogue was being supplemented – and would eventually be displaced – by a conception of knowledge as an organized structure that exists outside the mind as an inspectable

[3] *"Philosophia est comprehensio disciplinarum liberalium inferiorum: et alias dicitur encyclopaedia ... Comprehensio ilia spectatur in mente ut habitus, extra mentem ut systema."*

object. The externalization of cognitive order is palpable in Alsted's distinction: the *systema* is precisely what remains when the *habitus* is projected outward, detached from the knower, and made available to inspection by anyone capable of following its structure.

The shift from *habitus* to *systema* implied a shift from a world in which cognitive authority was vested in persons to a world in which it was vested in structures. This is the deepest consequence of Keckermann's innovation, and the one that ramifies most extensively through the subsequent history of European thought: once the locus of cognitive authority begins to migrate from the person to the structure, the structure can be asked to do progressively more. What begins as an instrument of transmission can become an instrument of certification; what begins as an alternative to institutional authority can become a model for a different kind of authority altogether.

For once doctrine is organized so that its coherence appears to rest in an articulated structure, authority itself begins to take on a new form. One can still appeal to Scripture, confession, and office; Keckermann remained thoroughly embedded in a confessional world. Yet the standing of the doctrine is now partly borne by the visible order of its presentation. The authority that had previously resided more unequivocally in institution and tradition is beginning, however imperfectly, to migrate into the structure of knowledge itself.

In Keckermann, that migration remains above all a pedagogical and doctrinal phenomenon. The *systema* is not yet a general theory of legitimacy, nor a modern account of publicly auditable reason. But a pattern has been established whose later significance is unmistakable. If a doctrine's order can be displayed in a structure available to all who learn it, then authority becomes, in part, answerable to articulated form. The groundwork is laid for later aspirations to impersonal accountability: for the thought that what binds the mind, or even what binds a collectivity, should do so because it can be seen to occupy the right place within a publicly graspable order.

The same point also yields the earliest precondition of the epistemological rationale. Keckermann's *systema* is not yet designed to secure certainty against skepticism; it presupposes, rather than manufactures, a body of teaching to be organized. But by making order increasingly internal to the arrangement of the discipline, it helps prepare the thought that correctness might be legible in form itself. Once knowledge is expected to carry its order within its articulated structure rather than resting wholly on external institutional guarantee, the path is open to later thinkers who will ask whether the right structure can itself certify the rightness of the content.

What drove Keckermann's innovation were thus pressing practical needs, and recognizing this transforms our understanding of what the concept of system originally meant. The *systema* was a technology devised in response to the loss of Catholic institutional authority and educational infrastructure. Without an established church hierarchy to enforce theological unity, Reformers had to devise internal structural criteria. The ideal of systematicity supplied those criteria in the form of consistency, coherence, completeness, and parsimonious principledness. The authority that had resided in an institution was relocated to a structure. And a structure, unlike an institution, could be inspected by anyone who understood its principles. As Timothy Watson's recent study puts it, Keckermann's innovation "gives the logician the directing role as a technical expert in mining the truth of theology, philosophy, and other sciences," a role that "flies in the face of the medieval view of the Church and its elders controlling that role" (Watson 2023, 15).

5. Printing, Paper Technologies, and the Material Ecology of System

The confessional crisis was only one of the pressures to which cognitive systematization formed an answer. A second pressure, operating simultaneously and reinforcing the first, was the sheer material explosion of available knowledge that the printing press had unleashed. What Keckermann achieved conceptually in the form of the textbook was paralleled, enabled, and in some respects anticipated by a wider set of paper technologies through which early modern Europe learned to handle the growing abundance of knowledge.

By the time Keckermann published his *Systema Logicae*, the printing revolution was already more than a century old, and its cumulative effects had become impossible to ignore. The rapid multiplication of printed texts had created something for which the older intellectual culture possessed no adequate response: a surfeit of information that exceeded any individual's capacity to absorb. The *multitudo librorum*, or multitude of books, was a matter of widespread comment and growing anxiety among the learned by the mid-sixteenth century (Blair 2010, 11–15, 55–61). The flood of new publications, many of dubious quality, was pulling readers away from the ancient authorities that humanist pedagogy cherished and overwhelming the topical methods of organization that had served the learned world tolerably well in the age of manuscripts.

The same technology that had made the Reformation possible – Luther's theses circulated at unprecedented speed thanks to print – thus also generated a problem of cognitive management that the Reformation's own pedagogical institutions then had to solve. Print broke up the Catholic monopoly on doctrinal dissemination by multiplying Protestant texts alongside Catholic ones. This produced an information environment in which it became increasingly challenging for any single reader to survey the whole. Confessional competition spurred publication, and the resulting abundance of competing claims generated an acute practical need for the reimposition of cognitive order and surveyability.

The connection to cognitive systematization is direct. It was as a reaction to this confusion about what followers of the new faith were supposed to believe, a confusion exacerbated by the printing press, that Protestant theologians began to publish synoptic expositions of the new theological doctrines and refer to them as “systems” (Ritschl 1906, 16). Information overload did not by itself generate the ideal of system. But it made forms of order that promised economy and navigability much more attractive. David Simpson (1993) and Clifford Siskin (2016) have developed this point by arguing that print and system were mutually reinforcing developments: cognitive systematization was a response to information overload, and systems also took hold thanks to the standardization and easy reproducibility that print introduced. The printed textbook, unlike the copied manuscript, was identical across all its instances; every student working from the same edition encountered the same definitions, the same divisions, the same ordered sequence. The cognitive architecture encoded in the textbook was mechanically reproducible and emancipated from the original systematizer: the printing press guaranteed that the system’s structure would survive transmission without distortion, as Elizabeth Eisenstein argued in her account of printing as an “agent of change” that introduced fixity and standardization into a previously fluid manuscript culture (Eisenstein 1979, I:113–126). If the *systema* was a machine for organizing thought, the printing press was the machine that replicated that machine across an entire educational infrastructure. The entanglement of systematization and mechanization was present from the very beginning, embedded in the material conditions of the *systema*’s production and dissemination.

Cognitive systematization was part of a broader emergence of technologies of knowledge management in the face of this deluge of information. Techniques for excerpting and recombining textual material also gained in importance. Indices, alphabetized dictionaries, crossreferenced encyclopaedias, and increasingly sophisticated methods of note-taking all represented attempts to impose order on an unruly abundance (Blair 2010, 137–144, 210–225). Knowledge was increasingly handled as material that could be segmented, sorted, and rearranged. In this respect, the material culture of paper helped foster a conception of cognition in which the organization of knowledge could be partially externalized and materially manipulated.

An emblematic example is the technique devised by the Swiss polymath Konrad Gessner. In his *Pandectarum sive Partitionum Universalium* (1548), the second volume of his monumental *Bibliotheca Universalis*, Gessner published what Markus Krajewski has identified as perhaps the earliest explicit description of a procedure for generating and managing knowledge through the decomposition of texts into mobile, rearrangeable units (Krajewski 2011, 3–4, 13–14). The algorithm was simple: one copies passages of importance onto a sheet of paper, using one line per idea; one cuts the sheet into slips with scissors; one arranges the slips into clusters; one subdivides the clusters as often as necessary; and one fixes or copies the final arrangement (Krajewski 2011, 13). Gessner organized the knowledge surveyed in his *Pandectae* under *loci communes* arranged in a twenty-one-class tree structure, with grammar first and theology last; the commonplaces were "complemented and enhanced according to individual need," generating an open-ended system of topical coordinates for the classification of everything one read (Krajewski 2011, 10–12).

What makes Gessner's technique significant is the operation it performed on knowledge itself: decomposition into discrete, uniform, mobile carriers that could be arranged and rearranged according to strict systems of order. Krajewski characterizes this as the defining principle of the "paper machine": "information is available on separate, uniform, and mobile carriers and can be further arranged and processed according to strict systems of order" (Krajewski 2011, 3). The logic at work here is the logic of conceptual mechanization: knowledge becomes divisible into units, rearrangeable by principle, surveyable in external form, retrievable independently of memory, and available to rule-governed handling.

This is the material counterpart of what Keckermann would achieve at the conceptual level a half-century later: breaking the continuous stream of knowledge into discrete units, organizing those units systematically, and making the resulting structure surveyable. The paper machine and the cognitive *systema* were twin responses to the same problem, operating on different levels, the one material, the other conceptual. Taken together, they reveal a broader ecology in which knowledge was coming to be treated as something that could be decomposed, externalized, and managed through structured procedure.

This broader ecology also helps explain why the pedagogical rationale for systematization quickly acquired wider resonance. A body of doctrine arranged as a *systema* possessed many of the same virtues that made indexes, tables, and bibliographies valuable. It shifted the work of organization outward: cognitive order became less wholly immanent to an accomplished mind and more visibly dependent on structures that could be built, inspected, and used. Seen in this light, Keckermann's *systema* belongs to a larger family of early modern devices for coping with complexity.

Together, the paper machine and the cognitive *systema* signal a transition towards the treatment of knowledge as something that could be rendered navigable through systematization. As Matthew Daniel Eddy has shown, it is the allure of these machine-like virtues that later incited Adam Smith to systematize the syllabus in logic and moral philosophy at Glasgow University, and it is no coincidence that this was happening in Calvinist Scotland: "the value assigned to the system as a developmental tool in Scotland and other Reformed educational contexts was in part an extension of the Calvinistic theological tradition that had shaped the learning experiences of students from the sixteenth century onward" (Eddy 2023, 271). Smith's efforts to systematize the teaching of philosophy underscore the durability of the connection between Reformed pedagogy and systematization.

6. The Metaphysical Warrant: First Appearance

An assumption embedded in Keckermann's enterprise that his own pedagogical arguments do little to justify is that the domain being systematized *admits* of systematization: that the disciplines possess an internal logical structure which the system merely makes explicit. Ritschl observed that early seventeenth-century theorists of *systema* operated on what he called a "pre-critical" assumption: the belief that all truths, whether evident or supernatural, exist objectively within or beyond the world, and that the knowing subject's task is simply to discover and correctly reproduce this objective order through proper method (Ritschl 1906, 49–50). For Keckermann and his immediate successors, this presupposition was underwritten by a theological conviction so deeply held as to require no argument: God's creation displays a rational order, and human knowledge, when properly organized, reflects that preexisting order.

Here the Stoic ancestry of the term matters again. When Keckermann borrowed *systēma* from cosmology and applied it to doctrine, he carried with him the Stoic assumption that the organization of knowledge should mirror the organization of the cosmos. The Stoic *systema mundi* was a rationally ordered totality governed by *logos*; Keckermann's *systema logicae* aspired to be a rationally ordered totality governed by logic. His conviction that logic and other disciplines can be systematized was grounded in the conviction that reality itself is systematic; the structure of a well-made textbook mirrors, at a remove, the structure of a well-made world.

If the notion of a philosophical "system" became thinkable through the analogy with Copernican world-systems, then what may be called the *metaphysical warrant* – the assumption that the systematization of thought is warranted by the antecedent systematicity of reality – was originally part of what supported the concept's extension from the ontological to the epistemological. The cosmological metaphor promised that the orderly arrangement of parts in a textbook was continuous with the orderly arrangement of parts in the world.

Yet to describe this assumption as tacit or unexamined, as though Keckermann simply assumed what he never articulated, would be to understate its role in his thought. Keckermann conceived of the relationship between logic, philosophy, and divine truth in terms of a series of images, in which each level of the cognitive hierarchy represents the one above it. Logic, he held, relates to philosophy "as an image represented in a mirror to the face which it represents," handling the "images and simulacra of philosophical things as instruments of knowing" (Keckermann, *Praecognita philosophica*, 111–12). And, "realized in encyclopaedic form, the whole of philosophy can be seen as an image of divine truth itself" (Burton 2024, 231–232). This is a version of the metaphysical warrant, one whose articulation draws on the Augustinian-Franciscan tradition of *exemplarism*, according to which all created structures participate in and reflect the divine ideas. Keckermann effectively treats the systematic organization of knowledge as mirroring the rational order of creation, which in turn mirrors the rational order of God's mind.

In fact, the Ramism that Keckermann builds on itself grew out of a long tradition of topical Realism – the philosophical belief that the categories and logical structures used to organize human knowledge actually correspond to the objective, divine structure of the universe. This tradition reached back through Rudolph Agricola and Philipp Melanchthon to medieval Neo-Platonic traditions, and it remained, as Simon Burton has recently argued in a comprehensive study, a fundamentally Platonist enterprise: the conviction that logical structure mirrors the intelligible order of reality, and hence that the divisions uncovered by dialectical analysis are divisions inscribed in things themselves (Burton 2024, 44–83). "The dialectic of the mind," Ramus maintained, was a light "emulating the eternal and blessed light" and the "image of the parent of all things" (Burton 2024, 56).

From the beginning, the early modern rise of the *systema* was thus accompanied by a justificatory intuition that would give the new ideal much of its later prestige: the thought that reality itself is ordered, and that a properly organized body of knowledge ought therefore to mirror that order. This intuition did not create the practical need for systematization. The pedagogical and media pressures described above were already sufficient to generate a demand for more surveyable, reproducible, and articulable forms of doctrine. But the metaphysical warrant elevated that practical technology into something possessing dignity as well as usefulness.

That dignity matters, because it helps explain the peculiar stability of the systematicity ideal once established. A merely convenient pedagogical format might have come and gone. The *systema* became something more durable because it could be defended on two levels at once. Practically, it solved urgent problems of teaching and transmission. Metaphysically, it could be presented as answerable to the very structure of the created world. The two levels were mutually reinforcing.

The form was useful because the world was orderly; the world's order, in turn, seemed to justify the form.

The metaphysical warrant should not be mistaken for the original spur to cognitive systematization. It did not, by itself, explain why *systema* emerged when and where it did. Belief in divine order long predated Keckermann. Medieval thinkers had also inhabited a created cosmos without calling their bodies of doctrine *systemata*. What was different in 1600 were the technological conditions and the acute pedagogical needs.

This priority of the pedagogical rationale matters for the argument of this article as a whole. Its central claim is that the ideal of systematicity arose not because philosophers first discovered a timeless truth about the essence of reason, but because historically situated actors needed ways of generating reliable cognitive order under pressure. The metaphysical warrant was powerful, and in some contexts indispensable, but it attached itself to an ideal whose practical utility had already become evident. God may have been invoked to justify systematization, but systematization first emerged in response to earthly problems.

Even so, once the warrant was in place, it transformed the horizon of the enterprise. A *systema* that merely transmitted doctrine efficiently remained a pedagogical device. A *systema* that mirrored the rational order of creation could aspire to more. It could become the visible expression of a world in which truth itself is systematic. And once that possibility had been glimpsed, the path opened toward the more ambitious epistemological projects of the seventeenth century: projects that would no longer be satisfied with a system because it teaches well, but would seek a system because only a properly ordered body of knowledge seems capable of securing certainty, completeness, and rational necessity.

The metaphysical warrant thus plays a transitional role. It does not originate the *systema*, but it helps prepare its expansion beyond pedagogy by providing assurances that the systematization of thought promotes *adaequatio intellectus ad rem* – the adequation of the intellect to things. The question of what happens when this assurance is withdrawn – of whether the demand for systematization can survive the collapse of its metaphysical warrant – is one of the most important questions in the subsequent history of philosophy.

7. A Historical Mirror: The Promise of Compressing Intellectual Labour

The transition from *Summa* to *systema* instantiates a pattern that recurs whenever new technologies of externalization promise to compress the labour of understanding. The *systema* offered to make the conclusions of extended inquiry available to learners who had not themselves undertaken the inquiry: the definitions stood ready, the divisions were laid out, and the ordered sequence guided the novice from principle to application. What the *Summa* had preserved, and what the *systema* compressed out, was the dialectical process through which those conclusions had been reached: the *quaestio*, the marshalling of objections, the *sed contra*, the painstaking accommodation of countervailing considerations in the *ad argumenta*. When a reader worked through the *Summa*, the path to the conclusion was itself part of what was understood. A Keckermann-style textbook could promise to compress a twelve-year formation into three years of study because it claimed to have found a shortcut to the destination.

Each subsequent technology of knowledge management has renewed this promise in its own idiom. The encyclopaedia, the card index, the searchable database, the algorithmically generated summary: each makes it easier to arrive at conclusions without traversing the terrain from which they were won. And each borrows, in its own register, the virtues that the *systema* first aspired to realize: efficiency, surveyability, and reproducibility. The machine-like character of systematized knowledge, figurative in Keckermann's textbook, became progressively less figurative as technologies have advanced. What has remained constant is the animating hope that the intellectual path can be shortened without significant loss, because what matters is the product of inquiry rather than the process by which it was forged.

The contrast between *Summa* and *systema* sharply throws into relief why that hope is often bound to be misguided. Knowledge is more than a collection of distinctions and propositions. It encompasses embodied intellectual dispositions, cultivated through sustained dialectical engagement, that shape the learner's capacity to notice things, to draw connections, to anticipate objections and weigh their force. True learning is formative as well as informative.

This was well understood by the medieval pedagogical tradition revolving around the *Summa* and the cultivation of *habitus*. Aquinas's *ad argumenta* forced the reader to confront the strongest available objections, and to appreciate why they carried weight, because he knew that this made a difference to what the reader came away with, and to their ability to hold on to tensions that could not fully be resolved. He also knew that the intellect is more than a repository waiting to be stacked with information. Sustained engagement with countervailing considerations over hours, months, or years leaves the mind altered in ways that a distillation of the results cannot replicate.

In part, this is because the connections a learner draws in the course of that engagement are shaped by the particular intellect doing the drawing: by what it already knows, by what it finds puzzling, by the unexpected adjacencies it discovers along the way. It is often those idiosyncratic connections that enable the true appropriation of knowledge.

By contrast, a system, whatever its material substrate, is a network of pre-drawn connections. This is what makes it so powerful as a technology of transmission. It delivers the same connections across every copy and every student. But this is also what limits its capacity to be formative. The very uniformity that enables a *systema* to teach at scale forecloses the idiosyncratic pathways that sustained engagement would have opened. The learner's network of connections becomes something received rather than something individually constructed.

The sixteenth-century transition thus illuminates a recurring structure. Whenever new means of externalization make knowledge appear increasingly detachable from the cognitive labour of the learner, people are tempted by the promise of a shortcut. But the shortcut is self-defeating when the journey is the point. No matter how perspicuously the supposed fruits of inquiry are displayed in the synopsis, the tension between efficiency and depth of engagement remains, and something threatens to be lost. What is lost thereby is harder to display, because it consists precisely in the kind of understanding that only the longer path could have cultivated: the *habitus* that Alsted distinguished from the *systema*, the disposition that takes shape in a particular mind through a particular encounter with the material, and that no externalized architecture can fully encode. The question whether understanding can be compressed without remainder into its products is as old as the *systema* itself. It has lost none of its force.

8. Conclusion

It cannot be denied that at some level, the pedagogical machine that was Keckermann's *systema* worked. Within a generation, the concept had migrated from the classrooms of the Danzig Gymnasium into the mainstream of European intellectual life, carried by the Calvinist pedagogical network and propelled by the printing press. The *systema* organized knowledge; it made knowledge surveyable; and it facilitated its transmission. It relocated cognitive authority from persons and institutions into articulated structure, and in doing so it gave thought a form whose functional virtues were the virtues of machinery: efficiency, surveyability, impersonality, and reproducibility. The pedagogical rationale explains why this first system arose. And the incipient relocation of authority from institution to structure prepared the conditions from which later, more ambitious rationales would emerge. What Keckermann's system did *not* do – what its inventor never asked it to do – was *certify* that the knowledge it organized was true. The Keckermann-style textbook assumed the reliability of the doctrines it systematized. It asked how best to arrange, compress, and transmit them.

This assumption was about to come under serious pressure. The same decades that saw the rise of the cognitive *systema* also saw the revival of an ancient philosophical challenge that struck at the very possibility of secure knowledge, a challenge that Keckermann's pedagogical machinery was wholly unequipped to answer. A system that organizes received opinion without certifying its foundations organizes something that may be worthless. The seventeenth century's philosophical innovation was to demand that the system do more: that it serve as an instrument for producing certainty, that the rational structure of a system be itself a guarantee of truth. The pedagogical machine was about to become an epistemological one. That transformation, and the resistance it provoked from the very beginning, belongs to a further chapter in the history of cognitive systematicity.

Bibliography

Arnim, Hans von, ed. Stoicorum Veterum Fragmenta. 4 vols. Stuttgart: Teubner, 1964. First published 1903–1924.

Blair, Ann M. Too Much to Know: Managing Scholarly Information before the Modern Age. New Haven: Yale University Press, 2010.

Burton, Simon J. G. Ramism and the Reformation of Method. Oxford: Oxford University Press, 2024.

Eddy, Matthew Daniel. Media and the Mind: Art, Science, and Notebooks as Paper Machines, 1700–1830. Chicago: University of Chicago Press, 2023.

Eisenstein, Elizabeth L. The Printing Press as an Agent of Change: Communications and Cultural Transformations in Early-Modern Europe. 2 vols. Cambridge: Cambridge University Press, 1979.

Freedman, Joseph S. "The Diffusion of the Writings of Petrus Ramus in Central Europe, c. 1570–c. 1630." Renaissance Quarterly 46, no. 1 (1993): 98–152.

Freedman, Joseph S. Philosophy and the Arts in Central Europe, 1500–1700: Teaching and Texts at Schools and Universities. Aldershot: Ashgate, 1999.

Hotson, Howard. Johann Heinrich Alsted, 1588–1638: Between Renaissance, Reformation, and Universal Reform. Oxford-Warburg Studies. Oxford: Clarendon Press, 2000.

Hotson, Howard. Commonplace Learning: Ramism and Its German Ramifications, 1543–1630. Oxford-Warburg Studies. Oxford: Oxford University Press, 2007.

Kenny, Anthony. Aquinas on Mind. Topics in Medieval Philosophy. London: Routledge, 1993.

Krajewski, Markus. Paper Machines: About Cards and Catalogs, 1548–1929. Translated by Peter Krapp. Cambridge, MA: MIT Press, 2011.

Lovejoy, Arthur O. The Great Chain of Being: A Study of the History of an Idea. Cambridge, MA: Harvard University Press, 1936.

Malcolm, Noel. "Thomas Harrison and His 'Ark of Studies': An Episode in the History of the Organization of Knowledge." The Seventeenth Century 19, no. 2 (2004): 196–232.

Muller, Richard A. Post-Reformation Reformed Dogmatics: The Rise and Development of Reformed Orthodoxy, ca. 1520 to ca. 1725. Vol. 1, Prolegomena to Theology. 2nd ed. Grand Rapids, MI: Baker Academic, 2003.

Ong, Walter J. "System, Space, and Intellect in Renaissance Symbolism." Bibliothèque d'Humanisme et Renaissance 18, no. 2 (1956): 222–239.

Ong, Walter J. Ramus, Method, and the Decay of Dialogue: From the Art of Discourse to the Art of Reason. Cambridge, MA: Harvard University Press, 1958.

Pasnau, Robert. Metaphysical Themes, 1274–1671. Oxford: Clarendon Press, 2011.

Rescher, Nicholas. "Historical Perspectives on the Systematization of Knowledge." In Epistemic Merit, 89–100. Heusenstamm: Ontos Verlag, 2006.

Ritschl, Otto. System und systematische Methode in der Geschichte des wissenschaftlichen Sprachgebrauchs und der philosophischen Methodologie. Bonn: Carl Georgi, 1906.

Secundant, Sergiy. "Formation of the Philosophical Concept of System: Bartholomäus Keckermann." Sententiae 34, no. 1 (2016a): 80–94.

Secundant, Sergiy. “Formation of the Philosophical Concept of System in Modern Time Philosophy: Clemens Timpler.” Sententiae 35, no. 2 (2016b): 41–56.

Siskin, Clifford. System: The Shaping of Modern Knowledge. Cambridge, MA: MIT Press, 2016.

Watson, Timothy. “Keckermann, System, and the Rise of the Subject.” Philosophy and Theology 35 (2023): 1–28.

Williams, Bernard. Ethics and the Limits of Philosophy. London: Fontana, 1985.